

Part 2: Advanced Intuition & Hardware Realization

Research Notes: Compressed Sensing Systems

1 The Paradigm Shift: From Compression to Sensing

Standard signal processing relies on the Shannon-Nyquist theorem (sampling at $2x$ bandwidth), whereas Compressed Sensing (CS) relies on the **Sparsity** of the signal.

1.1 1. The Classical Approach (Transform Coding / JPEG)

Philosophy: "Sample First, Compress Later."

1. **Acquisition:** We acquire the full signal $f \in \mathbb{R}^N$ (e.g., $N = 10^6$ pixels). This is expensive in terms of sensor power and bandwidth.
2. **Transformation:** We project f onto a sparsifying basis Ψ (e.g., DCT or Wavelet):

$$x = \Psi^T f$$

3. **Truncation:** We observe that x is K -sparse (only K large coefficients). We discard the $N - K$ small coefficients.
4. **Storage:** We store the K values.

Deep Insight: The inefficiency is obvious: We collected N samples of data only to discard $(N - K)$ of them immediately. The compression happens **after** the sensing cost has been paid.

1.2 2. The Compressed Sensing Approach

Philosophy: "Compress During Sensing."

1. **Measurement:** We do not measure pixels directly. We measure **linear projections** of the signal onto a measurement matrix Φ :

$$y = \Phi f$$

Here, $y \in \mathbb{R}^M$ where $M \ll N$. The compression happens **analogically** at the sensor.

2. **The Effective System:** If the signal f is sparse in basis Ψ (i.e., $f = \Psi x$), then:

$$y = \Phi(\Psi x) = Ax$$

where $A = \Phi\Psi$ is the "Holographic Dictionary."

3. **Reconstruction:** We recover x from y using non-linear optimization (ISTA/LASSO).

2 The Physical Reality of A : Hardware-in-the-Loop

The matrix A is not merely a data structure; it describes the physical interaction between the world and your sensor.

2.1 Case Study: The Single Pixel Camera (SPC)

In an SPC, we replace a megapixel sensor array with a single photodiode.

- **The Modulator:** A Digital Micromirror Device (DMD) acts as the physical implementation of Φ .
- **The Row ϕ_i :** At time step i , the DMD displays a random pattern of 1s and 0s. This pattern corresponds to the i -th row of the measurement matrix.
- **The Analog Dot Product:** The lens focuses light reflected from the DMD onto the single photodiode. The sensor physically computes the inner product:

$$y_i = \int_{\text{sensor}} (\text{Pattern}_i \cdot \text{Image}) dA = \langle \phi_i, f \rangle$$

- **Conclusion:** The matrix multiplication $y = \Phi f$ is performed by the **optics**, at the speed of light, with zero power consumption for computation.

3 The Reconstruction Brain: FPGA Implementation

While the *sensing* ($y = \Phi f$) is instant, the *recovery* (x from y) is computationally heavy. This is where the FPGA shines.

3.1 Why FPGA for LISTA?

The reconstruction algorithm (LISTA) consists of repeated layers of:

$$x^{t+1} = \mathcal{S}_\theta(W_e y + W_g x^t)$$

1. **Massive Parallelism:** The operations $W_e y$ and $W_g x^t$ are large Matrix-Vector Multiplications (MVM). An FPGA can perform thousands of Multiply-Accumulate (MAC) operations per clock cycle, unlike a CPU which processes them sequentially.
2. **Unrolled Architecture:** Since LISTA has a fixed number of layers (e.g., $T = 10$), we can physically layout 10 processing blocks on the chip, creating a continuous pipeline. Data flows in, travels through the 10 blocks, and flows out as a reconstructed image.
3. **Latency:** This pipelined architecture allows for real-time reconstruction, crucial for video applications where software solvers would be too slow.

Deep Insight: **The Full System Pipeline:**

- **Physics (Optics):** Compresses $N \rightarrow M$ instantly.
- **Sensor:** Digitizes only M values (Low Power).
- **FPGA:** Hallucinates the N pixels back from M values using geometry (LISTA).